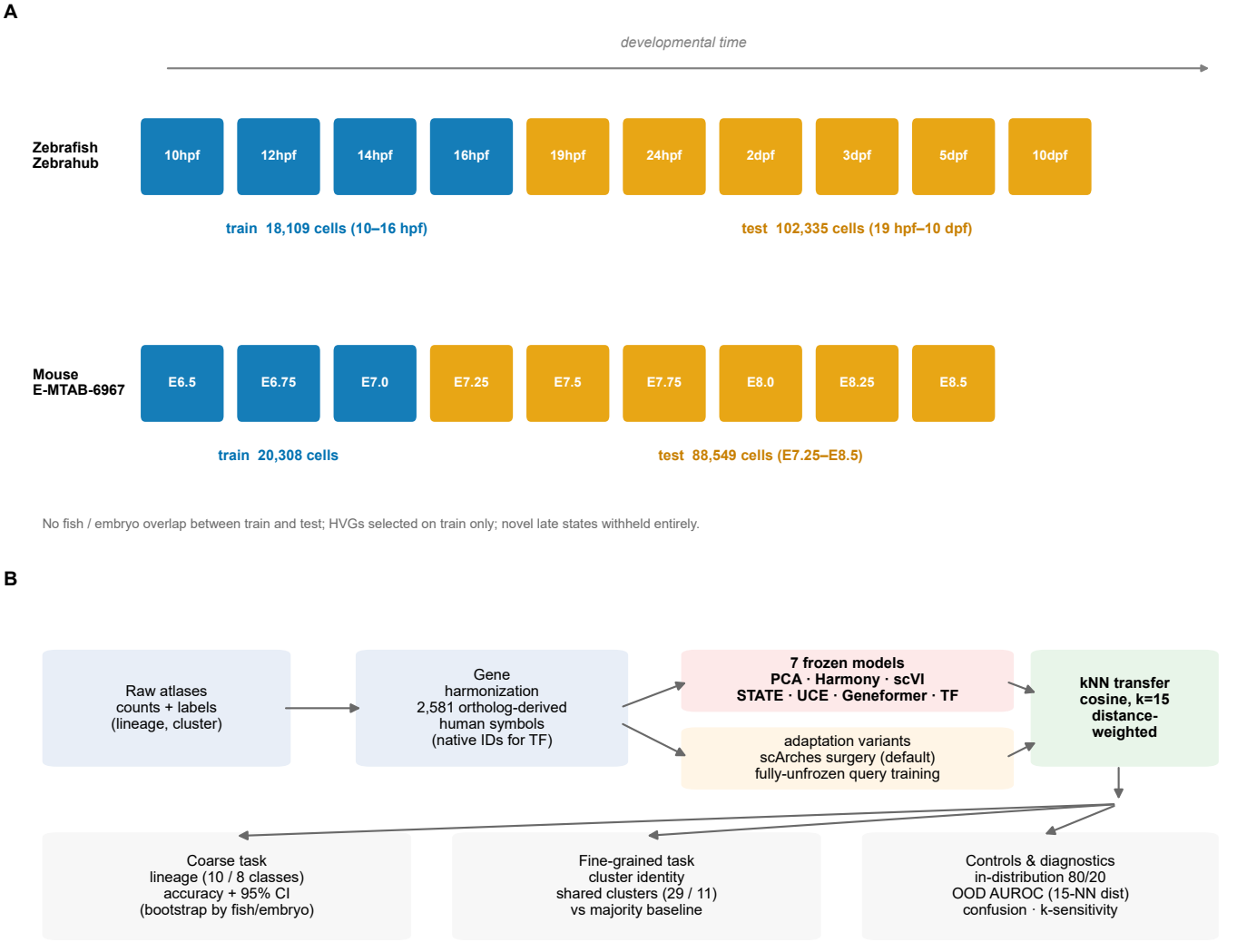

A
developmental time
Zebrafish
Zebrahub
10hpf
12hpf
14hpf
16hpf
19hpf
24hpf
2dpf
3dpf
5dpf
10dpf
train 18,109 cells (10–16 hpf)
test 102,335 cells (19 hpf–10 dpf)
Mouse
E-MTAB-6967
E6.5
E6.75
E7.0
E7.25
E7.5
E7.75
E8.0
E8.25
E8.5
train 20,308 cells
test 88,549 cells (E7.25–E8.5)
No fish / embryo overlap between train and test; HVGs selected on train only; novel late states withheld entirely.
B
Raw atlases
counts + labels
(lineage, cluster)
Gene
harmonization
2,581 ortholog-derived
human symbols
(native IDs for TF)
7 frozen models
PCA · Harmony · scVI
STATE · UCE · Geneformer · TF
adaptation variants
scArches surgery (default)
fully-unfrozen query training
kNN transfer
cosine, k=15
distance-
weighted
Coarse task
lineage (10 / 8 classes)
accuracy + 95% CI
(bootstrap by fish/embryo)
Fine-grained task
cluster identity
shared clusters (29 / 11)
vs majority baseline
Controls & diagnostics
in-distribution 80/20
OOD AUROC (15-NN dist)
confusion · k-sensitivity
Gene-set control: TF re-embedded on the corresponding 2,994 native genes
recoverable from the ortholog-derived set; robustness: k in {5,15,30}, 3 scVI seeds, unified 2,581-gene input.